

Instituto Tecnológico y de Estudios Superiores de Monterrey

Master of Applied Artificial Intelligence

SCOUT - Supply Chain Operational Vulnerability and Threat Assessment

Project Proposal and NDA

Team 44
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SCOUT Business and Data Understanding

This document describes the project context, the decisions SCOUT aims to support, and a preliminary data framework. The team has not yet received access to the MIT database. The proposed variables, labels, and evaluation sources therefore require validation with the sponsor before they can be treated as available data or as the basis for a trained model.

1 Background

The beneficiary institution is the Massachusetts Institute of Technology (MIT), through the Supply Chain Cybersecurity Lab at the MIT Center for Transportation and Logistics (MIT CTL). The Lab studies how cyberattacks affect physical supply chain operations, their technological dependencies, and organizations' ability to prevent, mitigate, and recover from disruptions [1, 2]. Dr. Camilo A. Mora-Quiñones guides the scope of SCOUT as the project's academic sponsor [1].

The research slide shared by the sponsor places three complementary tools within this agenda: MIT TRACE examines past attacks and their operational effects; MIT SCOUT assesses current conditions and translates technical vulnerabilities into priorities and recommendations; and MIT RADAR identifies emerging risks associated with technologies and trends [3]. This relationship describes the research agenda; it does not establish that TRACE or RADAR data have already been integrated into SCOUT.



The process affected by SCOUT is the assessment and prioritization of cyber risks in supply chain operations. A user identifies assets and dependencies that support physical processes, records known vulnerabilities or exposure conditions, and describes potential operational consequences. SCOUT then structures the evidence, determines explainable priorities, and presents options for mitigation, recovery, or further analysis. The intended product is an assessment prototype for the Lab's applied research and user testing; deployment in a specific company's production environment has not been established [1].

2 Business Understanding

Problem Formulation

SCOUT addresses a prioritization problem: a vulnerability severity score alone does not tell a supply chain team which exposed asset could disrupt a critical physical process, what the operational consequences might be, or which action deserves attention first. The prototype will bring technical evidence and process information into one reviewable assessment so teams can rank scenarios and explain the reasons for each priority [1, 4].

Context

Supply chains rely on shared IT and OT assets, suppliers, and linked physical processes, so a disruption can affect downstream operations. A severity score alone cannot account for local exposure, alternative capacity, recovery time, safety, or stoppage value. These factors help teams choose where to investigate and act. MIT CTL studies the operational effects of cyber incidents; SCOUT examines present conditions within that research agenda [2, 3]. Organizational loss size and frequency remain to be established.

Objectives

Design and demonstrate a prototype that combines vulnerability information with a guided self-assessment to produce a traceable operational risk profile, ranked scenarios, and contextual recommendations [1, 3]. The initial demonstration will define inputs and explicit rules, show dependencies on a value stream map when data allow, and test whether users can review the reasoning. Predictive performance requires separate evaluation on suitable real data [1].

Key Questions

- Which digital and control assets support each critical physical process, and how do sites, processes, and suppliers depend on one another?
- Which vulnerabilities or exposure conditions apply to those assets, and what evidence supports that assessment?
- How do process criticality, possible downtime, redundancy, recovery time, safety, and stoppage value change the priority of each scenario?
- Which mitigation, verification, or recovery actions fit each scenario, and who should review their feasibility?
- Which inputs are missing or uncertain, and when should SCOUT flag a scenario for human review?

Stakeholders

MIT CTL and Dr. Mora sponsor and review the scope; Team 44 designs and demonstrates the prototype. The table identifies proposed pilot roles, stages, and involvement. MIT must confirm external participants, approval responsibilities, and data access.

Stakeholder	Participation and Stage	Level of Involvement
MIT CTL and Dr. Mora	Scope definition, evidence access, review of criteria and results.	Academic sponsorship and expert validation.
Team 44	Questionnaire and data design, prototype construction, testing, and reporting.	Project execution.
Supply chain and operations professionals	Pilot input on processes, dependencies, continuity, and disruption impact; usability review.	Operational input and feedback; participants to be confirmed.
IT, OT, and cybersecurity personnel	Pilot checks of asset inventory, applicability of vulnerabilities, controls, and action feasibility.	Technical validation; participants to be confirmed.
Risk, continuity, and management teams	Review of business priorities, risk thresholds, and usefulness of suggested actions.	Decision input and feedback; participants to be confirmed.

3 Data Understanding

Access Status and Unit of Analysis

As of this submission, the team has not received the MIT database, a data dictionary, a sample size, access permissions, or verified labels. We do not claim that a training table exists, that the fields listed here are present in the database, or that record distributions, correlations, or quality have been measured. The proposed unit of analysis for the prototype is a scenario defined by a "vulnerability or exposure condition × asset × process" combination. When no CVE applies, an operational condition could be assessed under its own identifier. Multiple assets may support one process, and one asset may affect multiple processes.

Potential Sources and Contents

Source and Status	Useful Content	Limitation or Check
NVD and CVSS v4.0; potential public source [4, 5]	CVE, description, affected product, technical vector and score when available.	A published CVE does not prove a local asset is vulnerable; verify version and configuration.
CISA KEV; potential public source [6]	Indicator of observed exploitation for listed CVEs.	Absence from the catalog does not establish absence of exploitation.
FIRST EPSS; optional public source [7]	Estimated probability of exploitation of a CVE within 30 days, with query date.	An exploitation probability, not the probability of a stoppage or operational loss.
SCOUT self-assessment; to be designed and collected	Inventory, processes, dependency topology, exposure, redundancy, continuity, safety, and stoppage.	Self-reported answers require scales, supporting evidence, and expert review.
MIT Lab data; access pending	Potential incidents or historical evidence relevant to operations.	Schema, coverage, licenses, and possible links to SCOUT remain unconfirmed; do not assume labels.

The MIT slide describes TRACE as an observatory of attacks and their effects on supply chain operations [3], while the Lab's website confirms that a global database is being built [2]. At this stage, TRACE provides research context and is a potential future source, not a dataset available to the team.

Proposed Variables

Group	Candidate Inputs X	Record and Source
Technical	CVE ID; available CVSS v4.0 vector and score; product and version; KEV inclusion; EPSS and date.	Public sources and applicability checks by IT/OT.
Asset and exposure	IT/OT type; site; network reachability; external connectivity; function; compensating controls.	Inventory and self-assessment, subject to validation.
Operations	Affected process; criticality; supplier dependence; backup options; recovery time; safety implications.	Operations, continuity, and safety personnel.
Consequences	Estimated downtime and value per hour if available; downstream processes; alternative capacity.	Reported estimates, with ranges or "unknown."
Quality	Date, source, assessor, evidence, missing fields, and response confidence.	Metadata for audit and human review.

Business impact variables would remain separate from the official CVSS score: SCOUT can use CVSS as an input and construct its own operational priority without presenting that priority as an official CVSS score. Relationships among processes and assets could be used to color a value stream map if the topology can be documented. The color would represent the scenario's assigned category, not an already calibrated probability of disruption [1, 4].

Output and Machine Learning Technique

Immediate prototype output. An ordinal operational priority category (for example, critical, high, medium, or low), traceable reasons, suggested actions, and an "insufficient information / review required" flag. Thresholds, rules, and wording must be agreed with the sponsor. Initial prioritization will use an explicit rules-based baseline and controlled test cases. This baseline is not, by itself, a trained machine learning model [1].

Proposed ML technique for a data-supported phase. Supervised ordinal or multiclass classification: X would contain the technical and operational inputs above, and y would be the priority independently assigned by experts to assessed scenarios, using a documented rubric, date, and inter-rater agreement. A validated recommendation could also be learned, but that would require separate action labels and success criteria. The availability, number, and consistency of these labels are unknown. The team will not train a classifier or report accuracy, F1, or other classification metrics until sufficient labeled cases and an evaluation split are available that prevents the same incident or company from appearing in both training and test sets.

For the first prototype, synthetic scenarios and specialist review can test the coherence, traceability, and sensitivity of the rules, but cannot demonstrate predictive performance on real data. A category generated by those same rules must not be reused as a label to claim that a model has learned operational risk. If MIT provides historical records, the team will first review their fields, coverage, biases, permissions, and links among vulnerability, asset, and disruption; only then will the final ML technique be confirmed.

4 References

[1] Team 44. (2026, September 20). SCOUT Project Statement. Academic deliverable provided for this assignment.

[2] MIT Center for Transportation and Logistics. (n.d.). Supply Chain Cybersecurity Lab. https://ctl.mit.edu/research/supply-chain-cybersecurity-lab

[3] Rodríguez García, M., & Mora-Quiñones, C. (2026). Our Research, slide 18. Presentation material provided by the sponsor; unpublished in this submission.

[4] FIRST. (2024). Common Vulnerability Scoring System v4.0 Specification Document. https://www.first.org/cvss/v4.0/specification-document

[5] National Institute of Standards and Technology. (n.d.). National Vulnerability Database. https://nvd.nist.gov/

[6] Cybersecurity and Infrastructure Security Agency. (n.d.). Known Exploited Vulnerabilities Catalog. https://www.cisa.gov/known-exploited-vulnerabilities-catalog

[7] FIRST. (n.d.). Exploit Prediction Scoring System. https://www.first.org/epss/